The Role of Arbovirus UTRs on Neurotropism

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1. Background and outline

Emerging and re-emerging arthropod-borne viruses such as Japanese encephalitis (JEV), Dengue (DENV), Yellow fever (YFV), and Chikungunya (CHIKV) viruses are a growing global health threat. **Zika virus** (ZIKV) is a neurotropic flavivirus (FV) that can cause **congenital infection**, which can result in **microcephaly** and **fetal demise**. Recently, the translational regulator protein **Musashi-1** (Msi1) has been attributed to **promoting ZIKV replication, neurotropism, and pathology** [1]. Msi1 predominantly binds single-stranded `UAG` motifs in the 3'UTR of RNA [2].

Here we systematically analyzed the thermodynamic properties of **Musashi binding elements** (MBEs) in the 3'UTR of 76 arbovirus genomes *in silico*. Our results indicate that MBEs in the ZIKV 3'UTR occur predominantly in unpaired, single-stranded structural context, thus **supporting experimental observations** of Msi1 binding affinity with a thermodynamic model of RNA structure formation.

2. Flavivirus 3'UTR mediates pathogenicity

Flaviviruses are small (+)ssRNA viruses of 10-12kB length with highly structured UTRs. Upon infection, accumulation of stable long non-coding viral RNA, subgenomic flaviviral RNA (**sfRNA**), is observed. sfRNAs can modulate cellular function and are linked to pathogenicity. They are stable decay intermediates produced by partial degradation of the viral genome by 5'-3' exoribonuclease Xrn1, which is efficiently stalled at **evolutionary conserved xrRNA** structures.

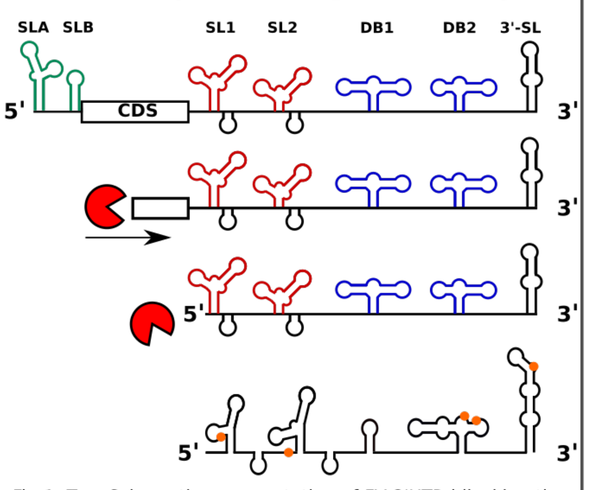


Fig 1. Top: Schematic representation of FV 3'UTR hijacking the host mRNA degradation pathway. Conserved xrRNA stem-loop (SL) and dumbbell (DB) elements are located in single or tandem copies within 3'UTRs and stall the host exonuclease Xrn1 (red pac-man). Bottom: ZIKV has two SL and one DB element. Location of UAG MBEs are highlighted in orange.

3. Opening energy and single-strandedness

We use the **ViennaRNA Package** [3] to model the thermodynamics of RNA secondary structure formation. The partition function Z allows for computation of the equilibrium probability of secondary structure s

$$Z = \sum_s e^{-E(s)/RT} \qquad P(s) = \frac{e^{-E(s)/RT}}{Z}$$

The accessibility (i.e., the probability that a region $i \ldots j$ along the RNA is single-stranded) can be derived from Z . The opening energy (i.e., the free energy required to force the region to be in a single-stranded structural context) is computed as $\Delta G_{\text{open}} = -RT \ln P(\text{unpaired})$.

Low opening energy indicates single-strandedness. We compute local pairing probabilities of trinucleotides to assess the likelihood of MBE single-strandedness in a genomic context. Comparison to a large sample of randomized sequences allows computing a z score for each trinucleotide.

$$z = \frac{\Delta G_{\text{open}}(x) - \mu}{\sigma}$$

4. MBE accessibility in ZIKV 3'UTR

We analyzed the accessibility of all trinucleotides in the coding region (CDS) and 3'UTR of ZIKV from Brazil and found a marked difference in the distribution of z scores, suggesting different sequence composition. Musashi-binding **`UAG` trinucleotides are maximally accessible** in the 3'UTR, which **corroborates experimental studies** [1,4].

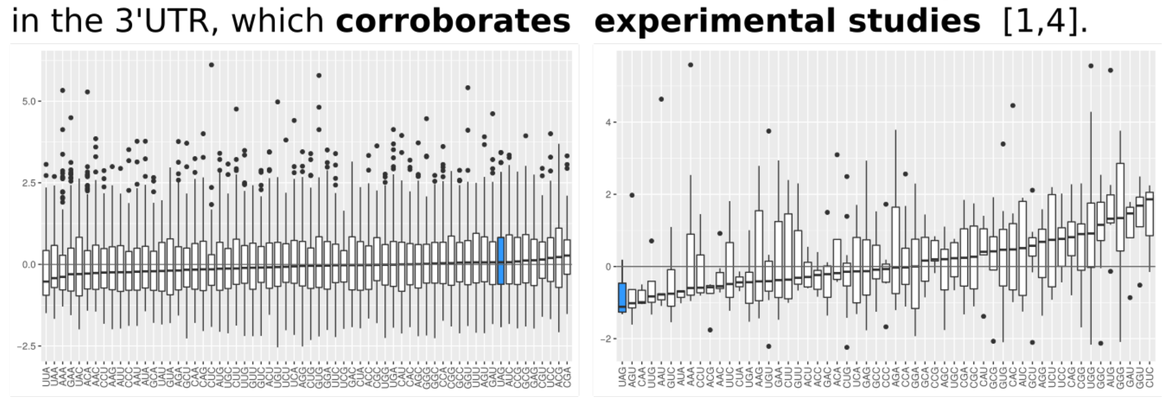


Fig 2. Distribution of z scores for all trinucleotides in the CDS (left) and 3'UTR (right) of ZIKV-BR, sorted by median z score. Interquartile ranges are homogeneous within the CDS region. The MBE motif UAG (blue) is maximally accessible in the 3'UTR.

5. MBE accessibility in the 3'UTR of flaviviruses and alphaviruses

To address the broader question whether other viruses have a similar neurotropic potential to ZIKV in the developing fetus, we analyzed MBEs in the 3'UTR of related arbovirus genomes. Below, we show the distribution of MBE opening energy z scores for flaviviruses (left) and alphaviruses (right). Low overall z scores indicate unpaired UAG motifs, suggesting **high Msi1 affinity**. We use this as an **estimator for teratogenicity**.

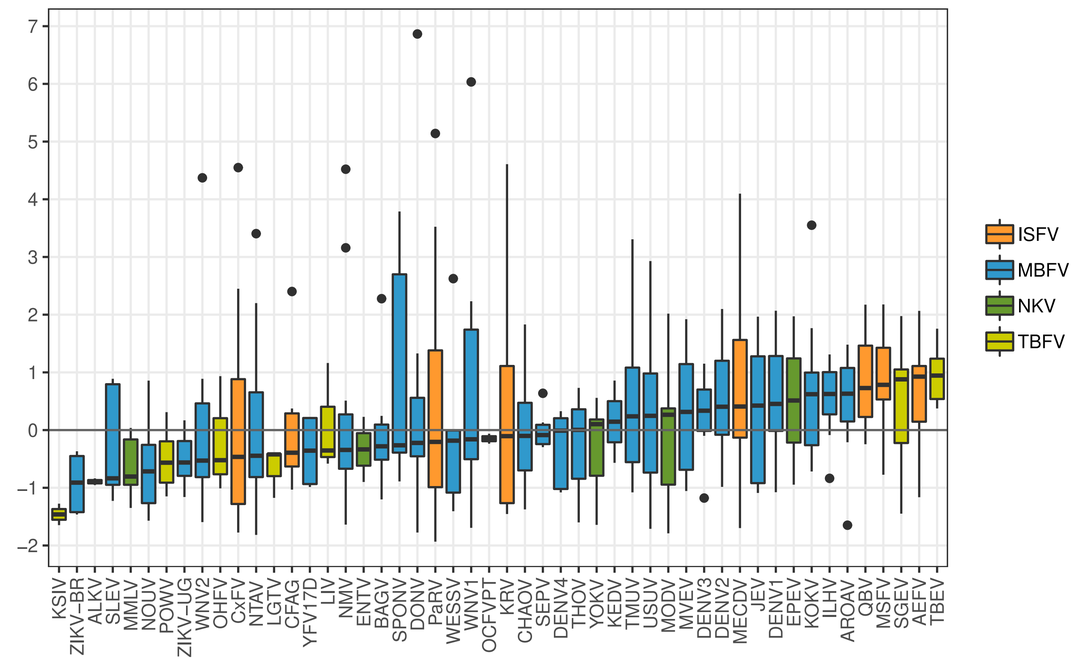


Fig 3. Distribution of MBE opening energy z scores in flavivirus 3'UTRs. Colors indicate sercomplexes (ISFV: Insect-specific flaviviruses; MBFV: Mosquito-borne flaviviruses; NKV: No known vector flaviviruses; TBFV: Tick-borne flaviviruses). Only viruses with more than two MBEs within the 3'UTR shown.

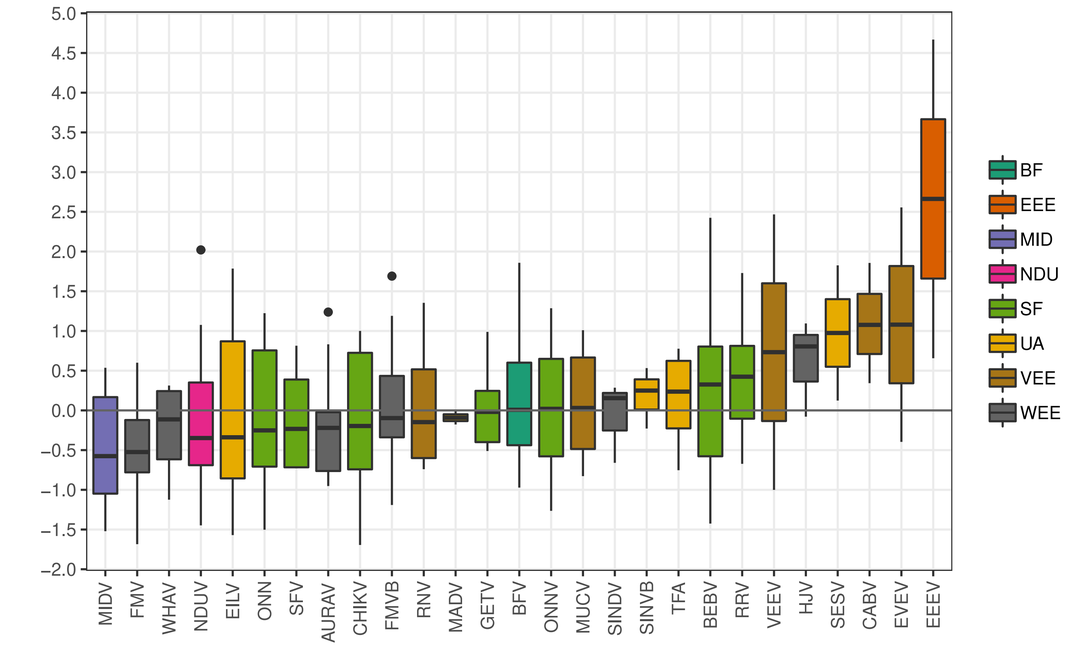


Fig.4. Distribution of MBE opening energy z score in alphavirus 3'UTRs. (BF: Barmah virus complex; EEE: Eastern equine encephalitis complex; MID: Middelburg virus complex; NDU: Ndumu virus complex; SF: Semliki forest complex; VEE: Venezuelan equine encephalitis complex; WEE: Western equine encephalitis complex; UA: Unassigned alphaviruses).

Among flaviviruses, the **Brazilian ZIKV has the lowest median MBE opening energy** z scores, followed by the neurotropic viruses SLEV, WNV, and POWV, which can cause transplacental infection, severe neuropathology and fetal demise [5]. Neurotropic alphaviruses such as SINDV, VEEV and EEEV, and the teratogenic SFV do not show a strictly negative z score distribution, as seen for some flaviviruses.

We employed an established biophysical model of RNA structure formation to analyze the thermodynamic properties of MBEs *in silico*. Our results underline experimental studies suggesting that **ZIKV is not alone in its capacity to cause severe neuropathology** [6]. The tropism of other arboviruses might have been overseen due to insufficient epidemiologic data.

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[1] Chavali, P.L. et al. (2017), Neurodevelopmental protein Musashi 1 interacts with the Zika genome and promotes viral replication, Science, eaam92433
[2] Uren, P.J. et al. (2015), RNA-binding protein Musashi1 is a central regulator of adhesion pathways in glioblastoma, Mol Cell Biol, 35.17, 2965-2978
[3] Lorenz, R. et al. (2011), ViennaRNA Package 2.0, Algorithm Mol Biol 6:1, 26
[4] Zearfoss, N.R. et al. (2014), A conserved three-nucleotide core motif defines Musashi RNA binding specificity, J Biol Chem 289(51):35530-35541
[5] Platt, D.J. et al. (2018), Zika virus-related neurotropic flaviviruses infect human placental explants and cause fetal demise in mice, Sci Transl Med, 10(426) eaao7090
[6] de Bernardi Schneider, A. et al. (2018) Musashi binding elements in Zika and related Flavivirus 3'UTRs: A comparative study *in silico*, bioRxiv 407833